25th National High School Physics Competition: Final Solutions

1.

1. Method 1: intercept at C on Ax, time t.

Let l be the $A-C$ distance and l_p the distance from B to C (Fig. 1). Then

$$l = vt \ , \ l_p = v_p t \qquad (1)$$

and

$$l_p^2 = d^2 + l^2 - 2dl\cos\theta . \qquad (2)$$

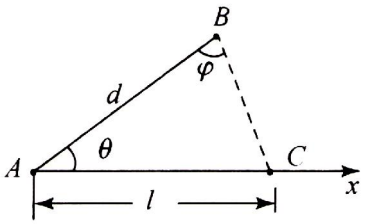


Fig. 1

Solving (1), (2) gives

$$l = \frac{d}{1-(v_p/v)^2}\{\cos\theta \pm [(\frac{v_p}{v})^2 - \sin^2\theta]^{1/2}\}. \qquad (3)$$

In (3), interception requires real roots, giving the necessary condition

$$v_p \geqslant v\sin\theta \ . \qquad (4)$$

Method 2: let φ be the angle between BA and BC (Fig. 1). The sine rule gives

$$\frac{l_p}{\sin\theta} = \frac{l}{\sin\varphi} \ .$$

Using (1) gives

$$\frac{v_p}{v} = \frac{\sin\theta}{\sin\varphi} \ .$$

$\sin\varphi \leqslant 1$ gives the necessary condition (4).

2. Let l_{min} be the least interception distance from A. From (3),

$$l_{min} = \frac{d}{1-(v_p/v)^2}\{\cos\theta \pm [(\frac{v_p}{v})^2 - \sin^2\theta]^{1/2}\}. \qquad (5)$$

If the receiver is closer to A than l_{min}, he gets the ball first. The condition for avoiding interception is

$$l_r < l_{min} \ . \qquad (6)$$

From (5), (6),

$$l_r < \frac{d}{1-(v_p/v)^2}\{\cos\theta \pm [(\frac{v_p}{v})^2 - \sin^2\theta]^{1/2}\} \qquad (7)$$

By (7), if a receiver on Ax, distance l_r from A, satisfies this inequality, the defender initially at B cannot intercept the ball.

3. Method 1: before the defender at B reaches C_1 on Ax, distance l_{min} from A,